

Light-Computing-Heat-Electricity-Water Multi-Generation System: A Full-Spectrum Spectral Splitting and Integrating Sphere Photovoltaic-Based All-Factor Clean Energy Complex for the AI Era

——Theoretical Architecture, Engineering Validation, and Multi-Scenario Deployment Strategy

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Abstract

The explosive growth of artificial intelligence has triggered an exponential surge in data center electricity demand and the wasteful discharge of massive quantities of low-grade waste heat, while conventional photovoltaic power generation remains persistently constrained by physical bottlenecks such as dust accumulation, high-temperature degradation, and large land-area requirements. This paper proposes and systematically demonstrates a "Light-Computing-Heat-Electricity-Water" five-element coupled all-factor clean energy complex. The system begins with spectral splitting technology to separate the full solar spectrum into visible and infrared pathways: visible light is transmitted via optical fibers into an indoor full-interior-wall photovoltaic integrating sphere power generation unit, leveraging the optical integrating sphere principle to achieve near-99% photon capture efficiency, with expected power output 15%–25% higher than an equal-area flat-mounted PV array. The infrared band drives a supercritical CO₂ (sCO₂) Brayton cycle for high-temperature power generation, achieving a cycle efficiency of up to 49% at a turbine inlet temperature of 700°C. The liquid-cooling return water from AI data centers is innovatively introduced as a medium-temperature preheating heat source, injected into

a Kalina cycle to realize an energy closed loop of "computing power is thermal power," with an expected waste heat recovery capacity of 5%–12% of total data center power consumption. In the low-temperature section, vortex tube grade restructuring and radiative cooling films are combined to achieve zero-energy-consumption cooling and residual electricity recovery. For coastal scenarios, ocean thermal energy conversion (OTEC) base-load power generation and low-temperature multi-effect distillation (LT-MED) seawater desalination are integrated, ultimately constructing a temperature-cascaded, multi-product co-generation energy self-sufficient system. Theoretical analysis indicates that the full-chain light-to-electricity conversion efficiency can exceed 37%, and when computational waste heat recovery and multi-product synergy are included, the comprehensive primary energy utilization rate can reach over 80%. This paper provides an in-depth demonstration from the perspectives of physical principles, thermodynamic architecture, full-chain efficiency modeling, key technology engineering validation, AI data center synergy mechanisms, multi-scenario deployment strategies, and full life-cycle economic analysis, and includes dedicated sections for critical discussion of system limitations, alternative technology pathways, and future research directions, aiming to provide a theoretical paradigm and engineering blueprint for the next generation of computing-energy converged infrastructure.

Keywords: Spectral splitting; Integrating sphere photovoltaics; Supercritical CO₂ Brayton cycle; Kalina cycle; AI data center waste heat recovery; Vortex tube; Radiative cooling; Ocean thermal energy conversion; Seawater desalination; Polygeneration system; Exergy analysis

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1. Introduction

1.1 From Energy Crisis to Computing Crisis: The Dual Challenge of the 21st Century

Human civilization in the 21st century faces an unprecedented dual challenge. On one hand, the unsustainability of fossil fuels has formed a global consensus—the Paris Agreement's 1.5°C temperature control target requires the world to achieve net-zero

emissions by 2050, necessitating a fundamental restructuring of the energy system. On the other hand, general artificial intelligence technologies represented by large language models are penetrating every corner of the socio-economic landscape at an unexpectedly rapid pace, and the computing infrastructure behind them is becoming one of the fastest-growing electricity-consuming sectors globally.

According to the International Energy Agency, global data center annual electricity consumption reached approximately 200–250 TWh in 2022, accounting for about 1%–1.3% of global electricity demand [1]. This figure is expected to grow to 260–360 TWh by 2026, and by 2030, driven by the exponential growth of AI large-model training and inference demand, global data center electricity consumption could exceed 1,000 TWh—equivalent to the annual electricity consumption of Japan. In China, the "East Data, West Computing" project has planned eight national computing hub nodes, targeting total national data center computing power exceeding 300 EFLOPS by 2025 [2].

Data center energy efficiency is typically measured by Power Usage Effectiveness (PUE), defined as the ratio of total data center input electricity to IT equipment actual electricity consumption, with an ideal value of 1.0. Over the past fifteen years, the industry PUE has experienced a significant decline from above 2.0 to nearly 1.1—advanced data centers of technology giants such as Google and Microsoft have achieved industry-leading PUEs of approximately 1.10–1.15. However, continuous PUE improvement has entered a regime of diminishing returns: the energy consumption of non-IT components such as cooling systems and power distribution systems has been compressed from over 50% to 10%–15% of total energy consumption. This means that even if PUE is further reduced to 1.01, data centers will still emit 30%–50% of their total electrical energy as low-grade waste heat.

The thermodynamic characteristics of this waste heat merit careful examination. The rapid proliferation of liquid cooling technologies (including cold plate, immersion, and spray cooling) has significantly elevated the grade of data center waste heat from the traditional 25–30°C (air-cooled exhaust) to 30–50°C (liquid-cooled return water). Haier's "Wind-Liquid Homology" solution has validated that the secondary side of liquid cooling can stably output 30–50°C high-quality waste heat [3], and the Lingang intelligent computing center's liquid cooling return water temperature reaches 35°C [4]. This opens new possibilities for waste heat utilization. However, the 30–50°C temperature range is neither sufficient to drive

traditional steam Rankine cycles (typically requiring $>100^{\circ}\text{C}$ heat sources) nor high enough for efficient absorption refrigeration cycles, leaving conventional utilization pathways severely constrained.

1.2 Technical Bottlenecks of Photovoltaic Power Generation: A Comprehensive Examination from Materials to Systems

On the clean energy supply side, photovoltaic power generation, as the fastest-growing renewable energy technology with cumulative installed capacity exceeding 1 TW (2022), has achieved a levelized cost of electricity lower than fossil fuels in many regions. However, ground-mounted PV power stations still face three major problems in actual operation, severely constraining their full life-cycle power generation efficiency and reliability.

Dust accumulation. PV modules are exposed to outdoor air for extended periods, and dust deposition is the most common factor causing efficiency decline. A review by Sarver et al. indicates that dust-induced PV efficiency losses range from 7.8%–19.2%, and can exceed 40% in dust storm-prone regions such as the Middle East and North Africa, and even reach 64% under certain dust types like coal dust [5]. When dust accumulation density reaches 10 g/m^2 , module output power drops by approximately 34% [6]. Large PV stations in Northwest China (Qinghai, Gansu, Ningxia, etc.) lose about 8%–15% of annual generation due to dust, with losses exceeding 20% during spring sandstorm seasons.

High-temperature degradation. PV cell output power decreases significantly with increasing temperature, with a temperature coefficient of approximately -0.3% to $-0.5\%/^{\circ}\text{C}$. That is, for every 1°C increase in cell temperature, output power drops by about 0.4%–0.5%. Under summer midday conditions, outdoor PV panels can reach operating temperatures of $60\text{--}80^{\circ}\text{C}$, which is $35\text{--}55^{\circ}\text{C}$ above standard test conditions (25°C), corresponding to power losses of approximately 14%–28%. More critically, in concentrated photovoltaic (CPV) systems with concentration ratios of tens to hundreds, without effective heat dissipation, cell temperatures can rise above 150°C within seconds, potentially causing permanent damage.

Large land area and O&M costs. Ground-mounted PV stations have a power density of approximately $150\text{--}200\text{ W/m}^2$ (based on 22% module efficiency and 85%

system efficiency), meaning a 1 GW station requires about 5–7 km² of land. In land-scarce regions (such as China's southeast coast, Europe, and Japan), land cost has become a significant constraint on PV project economics. In terms of O&M, beyond dust cleaning, detection and repair of issues such as micro-cracks, junction box failures, and inverter faults require sustained investment in professional teams and equipment.

1.3 Academic History Review of Data Center Waste Heat Utilization

Converting low-grade waste heat from data centers into useful energy has been a gradually emerging research direction over the past two decades. Based on the evolution of technical routes and system integration depth, research in this field can be divided into four generations.

First Generation (2005–2012): Direct air-cooled exhaust. Data centers in this period primarily used computer room air conditioning (CRAC) to cool and recirculate the hot air exhausted from servers, with PUE typically >1.8. Waste heat was not utilized in any way and was directly exhausted outdoors. A small number of studies began to assess the volume and utilization potential of data center waste heat resources [9].

Second Generation (2012–2018): District heating utilization. As data center scale expanded and liquid cooling technology emerged, researchers began exploring the injection of data center waste heat into district heating networks. Wahlroos et al. [10] analyzed the energy efficiency and economic benefits of supplying waste heat from a Finnish data center to the Helsinki district heating network, finding significant economic advantages when the data center is located near the heating network. However, the limitations of the district heating pathway are equally apparent: heating demand exhibits strong seasonal fluctuations (nearly zero in summer) and is geographically constrained by pipeline network coverage, lacking replicability.

Third Generation (2018–2023): Independent ORC recovery. The development of Organic Rankine Cycle (ORC) technology provided a new pathway for low-temperature waste heat power generation. Araya et al. [11] built a laboratory-scale ORC system using simulated data center 30–50°C waste heat as the heat source and R245fa as the working fluid, achieving approximately 5%–8% thermal-to-electric conversion efficiency. Ebrahimi et al. [12] proposed an optimal design method for data center ORC waste heat recovery systems. However, all schemes in this phase were

independent ORC systems utilizing only a single waste heat source, with neither efficiency nor economics reaching the threshold for commercial promotion.

Fourth Generation (2023–present): Multi-energy complementarity and cascade utilization. The breakthrough came from the introduction of solar-assisted approaches. Schaefer and Liaqat [13] (2025) provided the first systematic validation of a technical scheme using flat-plate solar collectors to preheat data center coolant before injecting it into an ORC system. Simulation results showed that solar assistance increased ORC power generation efficiency by over 8% during peak solar hours, with annual electricity generation increasing by 81% in Los Angeles and 60% in Ashburn, and levelized cost of electricity reduced by 16.5% and 5.5%, respectively. The study also revealed a counter-intuitive finding: for a fixed solar collector area (e.g., 300 kWth), the change in annual power generation between the lowest and highest solar resource locations was only about 25%, suggesting that this technology has broader geographical applicability than previously assumed—its economic viability depends more on local electricity prices and data center cooling load characteristics than on solar resource quality alone.

Although Schaefer and Liaqat's work represents a critical step forward, their scheme still has the following knowledge gaps:

(1) Only flat-plate collectors were used, without achieving spectral splitting of the solar spectrum—PV power generation and thermal power generation remain independent rather than synergistic.

The above knowledge gaps constitute the core research motivation of this paper.

1.4 Novel Contributions and Structural Organization of the Paper

The core novel contributions of this paper can be summarized as the following five innovations:

Innovation 1: First proposal of the "full-interior-wall PV integrating sphere" power generation unit concept, combining the ultimate photon capture characteristics of optical integrating spheres (theoretically >99%) with photovoltaic power generation, expected to achieve 15%–25% higher power output than an equal-area flat-mounted PV array.

Innovation 2: Construction of an "sCO₂ Brayton cycle + Kalina cycle" full-temperature-range series thermoelectric architecture, achieving complete cascade utilization coverage from 700°C high temperature down to 80°C low temperature.

Innovation 3: Reconfiguring the AI data center from a power consumer to a thermal contribution unit, realizing the paradigm shift of "computing power equals thermal power."

Innovation 4: Introducing vortex tube grade restructuring and radiative cooling films to form a "zero-energy-consumption cold skeleton," achieving end-stage cooling and grade upgrading without additional power consumption.

Innovation 5: Integrating OTEC base-load power generation with seawater desalination and mineral extraction for coastal scenarios, constructing an all-factor energy self-sufficient system "from sunlight and seawater to everything."

The physical boundaries and assumption conditions of this paper are declared as follows:

The light collection system is located in open space above ground, assuming annual average Direct Normal Irradiance (DNI) ≥ 1800 kWh/m², applicable to global mid- and low-latitude regions (between 40°N and 40°S).

The AI data center operates continuously (8,760 hours/year) with an average load rate of 70%–80%.

Coastal scenarios assume access to deep cold seawater at ≥ 100 m depth (4–10°C).

Economic analysis is based on 2026 Chinese market prices and policy environment, with carbon prices analyzed under three scenarios: 50–200 CNY/tCO₂.

Large-scale seasonal thermal storage (such as inter-seasonal thermal reservoirs) is not considered; the system focuses on daytime operation optimization.

Paper structure: Chapter 2 presents the complete thermodynamic architecture and energy flow topology of the system; Chapter 3 provides in-depth engineering validation of six key technologies; Chapter 4 establishes a full-chain efficiency model and exergy analysis framework; Chapter 5 analyzes the thermodynamic coupling and dynamic synergy mechanism between the AI data center and the system; Chapter 6 discusses deployment

strategies for three representative scenarios: tropical remote islands, coastal AI clusters, and large industrial complexes; Chapter 7 performs a full life-cycle economic analysis; Chapter 8 provides critical discussion of system limitations, alternative technology pathways, and future research directions; and Chapter 9 summarizes the conclusions.

2. System Thermodynamic Architecture

2.1 Physical Foundations of the Design Philosophy

The core design philosophy of this system can be distilled into the twelve-character principle: "divide and conquer, cascade utilization, closed-loop circulation." Its physical foundation is deeply rooted in the energy distribution characteristics of the solar spectrum and the spectral response characteristics of photovoltaic cells.

Solar radiation received at the Earth's surface (AM1.5G standard spectrum) has an integrated irradiance of 1,000 W/m² across the 0.3–2.5 μm wavelength band. The visible band (400–700 nm) accounts for approximately 43% of total energy, the infrared band (>700 nm) accounts for approximately 52%, and the ultraviolet band (<400 nm) accounts for approximately 5%. The spectral response range of mainstream silicon-based PV cells covers approximately 300–1,100 nm, with peak quantum efficiency in the 600–900 nm band. Photons with energy below the silicon bandgap (1.12 eV, corresponding to ~1,100 nm) cannot excite electron-hole pairs and are dissipated as heat; photons with energy significantly above the bandgap generate excess kinetic energy through the thermalization effect, which also ultimately converts into heat. Therefore, in conventional unfiltered concentrated PV systems, over 50% of the incident solar energy is converted into heat rather than electricity.

The fundamental idea of spectral splitting technology is to separate different bands of radiation at the front end of the optical path and direct them to the conversion terminals best suited to their energy characteristics. This can be described by the following mathematical expression:

where η_{PV} is the photovoltaic conversion efficiency (related to cell temperature T_{cell} and wavelength λ), and η_{thermal} is the heat engine cycle efficiency (related to heat source temperature T_{source}). When visible and infrared light are separated, the PV cells no longer bear the infrared thermal load, temperature rise is significantly reduced, and efficiency is maintained near the design point. Meanwhile, the infrared heat is no longer "wasted" on

heating PV cells but is instead directed to a high-temperature heat engine where it can be more efficiently converted to electricity.

2.2 Thermodynamic Principles of Full-Temperature-Range Cascade Utilization

The second law of thermodynamics dictates that no single heat engine can maintain high efficiency across a wide temperature range. Carnot efficiency $\eta_{\text{Carnot}} = 1 - T_{\text{L}}/T_{\text{H}}$, where T_{L} is the cold source temperature and T_{H} is the heat source temperature. For a 700°C (973 K) heat source with a 25°C (298 K) ambient cold source, the Carnot efficiency can reach 69.4%. However, if a single heat engine is required to handle a temperature span from 700°C to 30°C, the extreme irreversible losses in the heat exchange process significantly reduce the actual thermal efficiency.

This system employs a three-stage heat engine series architecture, with each stage's operating temperature range optimally matched to the thermodynamic characteristics of its internal cycle:

First stage—sCO₂ Brayton cycle: Heat source temperature 550–700°C, cold source temperature approximately 250–350°C (exhaust heat injected into the next stage), operating temperature span 300–400°C. sCO₂ near its critical point (31°C, 7.38 MPa) exhibits very high density (approximately 468 kg/m³) and low compression work, making the recompression cycle architecture particularly efficient. Compared with conventional steam Rankine cycles at the same temperature level, sCO₂ cycles can improve thermal efficiency by 3%–5% [14,15], while the turbomachinery is more compact—a 10 MW sCO₂ turbine is approximately 1/10 the volume of a steam turbine of equivalent power.

Second stage—Kalina cycle: Heat source temperature 80–350°C, cold source temperature approximately 30–80°C (depending on cooling conditions). The Kalina cycle uses an ammonia-water mixture as the working fluid, whose non-isothermal evaporation characteristics are highly matched to the cooling curves of medium-low temperature multi-source combined heat sources, effectively reducing pinch point temperature differences and exergy losses during heat exchange [16].

Third stage—Vortex tube + radiative cooling: Operating temperature range <80°C. In this temperature range, the efficiency of conventional heat engines approaches zero (Carnot efficiency <15%, actual efficiency <5%). Non-heat-engine

methods are employed (vortex tube grade restructuring and passive radiative cooling) to achieve the final "clean-up" recovery of energy.

This multi-stage architecture follows the thermodynamic exergy analysis principle of "temperature-matched, cascade utilization," maximizing the exergy efficiency of the entire system.

2.3 Detailed Energy Flow Topology

The complete energy flow topology of the system is shown in Figure 1 (note: an energy flow diagram should be included for formal publication). The temperature, energy, and exergy efficiency at each key node are annotated in detail below.

Node ①—Roof light collection and splitting stage: Total incident solar radiation 100 units. After concentration by Fresnel lens/mirror arrays (optical efficiency $\sim 90\%$), the light is split into two paths. The visible light channel (~ 43 units) is reflected by a dichroic beam splitter into the optical fiber network; the infrared channel (~ 57 units) passes through the splitter into the high-temperature heat receiver. The splitter's cutoff wavelength is set at 700 nm—wavelengths >700 nm enter the thermal pathway, wavelengths <700 nm enter the photovoltaic pathway.

Node ②—High-temperature heat collection and sCO₂ cycle inlet: Infrared radiation ($57 \text{ units} \times 90\% \text{ collector efficiency} \approx 51.3 \text{ units}$) heats sCO₂ working fluid from approximately 250°C to 550–700°C (depending on concentration ratio and thermal storage design). sCO₂ enters the turbine for expansion power generation, with a cycle efficiency of 44% (conservative value for 700°C inlet, 250°C outlet). Power output: $51.3 \times 44\% \approx 22.57 \text{ units}$. sCO₂ exhaust afterheat ($51.3 - 22.57 = 28.73 \text{ units}$) enters the Kalina cycle evaporator as the primary heat source for the next stage.

Node ③—Visible light transmission and integrating sphere PV: Visible light ($43 \text{ units} \times 90\% \text{ splitting efficiency} \times 90\% \text{ fiber transmission efficiency} \approx 34.83 \text{ units}$) is introduced into the integrating sphere through the light inlet (aperture ratio $<3\%$). After multiple reflections inside the sphere, the photon capture rate $>95\%$, with actual light energy absorbed by the PV cells approximately 33.09 units. Cell efficiency at 23% (accounting for temperature effect, with cell temperature maintained at 35–55°C via active cooling): power output $33.09 \times 25\% \approx 8.27 \text{ units}$. PV cell heat (generated by the 75% of absorbed energy not converted to electricity, approximately $33.09 \times 75\% \approx$

24.82 units) is carried away by the cooling air, of which approximately 60% (14.89 units) is recoverable at 70–130°C and enters the Kalina cycle.

Node ④—Kalina cycle multi-source confluence: The Kalina evaporator receives three heat sources: (a) sCO₂ exhaust afterheat 28.73 units (250°C→150°C cooling section); (b) integrating sphere PV hot air 15.94 units (70–130°C); (c) AI data center liquid cooling return water 9 MW (based on 22.5 MW data center, 40% waste heat ratio, normalized to approximately 8.96 units). Total heat input ≈ 53.67 units. Kalina cycle efficiency 12% (conservative value for multi-source confluence): power output $53.67 \times 12\% \approx 6.44$ units.

Node ⑤—Vortex tube grade restructuring: Low-grade waste heat from the Kalina cycle exit (below approximately 80°C), together with unrecovered low-temperature exhaust from the sphere, enters the vortex tube system. Using sCO₂ cycle bleed air or pressurized hot air as the compressed gas source, the vortex tube separates the airflow into a hot end (80–120°C, fed back to Kalina feedwater preheating, equivalent to approximately 0.5 units recovered) and a cold end (20–30°C below ambient, used for data center inlet air pre-cooling, equivalent to approximately 0.3 units of cooling electricity saved). Estimated total equivalent recovered electricity: 0.8 units.

Node ⑥—OTEC base load and seawater desalination: Deep cold seawater (4–10°C) serves as the ultimate cold sink, simultaneously providing cooling for the sCO₂ cycle pre-cooler, Kalina cycle condenser, and OTEC evaporator. OTEC uses the temperature difference between 20–30°C surface seawater and 4–10°C deep cold seawater for power generation, delivering 24-hour continuous base-load electricity. Low-temperature multi-effect distillation (LT-MED) utilizes the OTEC discharge seawater temperature difference to produce fresh water at approximately 311 t/h (based on 10 MW system scale, specific data in Chapter 6).

Node ⑦—Net output summary: PV power generation 8.27 + sCO₂ power generation 22.57 + Kalina power generation 6.44 + vortex tube recovery 0.8 + OTEC base load (configuration-dependent) ≈ 38.08 units. After deducting system parasitic loads (pump work, compression work, tracker power consumption, etc., approximately 15% of total output, about 5.71 units), the net output is approximately 32.37 units, corresponding to a full-chain net efficiency of approximately 32.4%.

The key parameters of each node are summarized in Table 1.

Table 1 Summary of Key Node Energy Flow Parameters of the System

Node	Temperature Range	Energy (Units)	Exergy Efficiency (Est.)	Primary Equipment
① Splitting Stage	Ambient	100 (Input)	—	Fresnel Lens + Dichroic Beam Splitter
② sCO ₂ High-Temp Stage	550–700°C	51.3 (Input)	~44%	sCO ₂ Turbine + Recuperator + Precooler
③ Integrating Sphere PV	35–55°C	34.83 (Input)	~25%	Sphere + PV Cells + Heat Sink
④ Kalina Medium-Temp Stage	80–350°C	53.67 (Three-Source)	~12%	Ammonia-Water Evaporator + Turbine + Condenser
⑤ Vortex Tube	<80°C	0.8 (Recovered)	~5–10%	Vortex Tube + Regulating Valve
⑥ OTEC Base Load	4–30°C (ΔT)	8 (Config-Dependent)	~2.5%	OTEC Heat Exchanger + Turbine + LT-MED
⑦ Net Output	—	~32.37	~32.4%	—

3. Key Technology Engineering Validation

3.1 Spectral Splitting and Light-Guiding Technology

Spectral splitting technology is the prerequisite for achieving full-spectrum cascade utilization. In essence, it performs "energy flow separation" at the front end of the optical path, directing different bands of solar radiation to the conversion terminals best suited to their energy characteristics. Based on the physical principles of splitting mechanisms, existing technologies can be divided into four categories [18–20]:

(1) Absorption-type splitting: Utilizes the selective absorption characteristics of liquids or solids to filter out specific bands. Nanofluid spectral splitting has been a research hotspot in recent years—metal or metal oxide nanoparticles are dispersed in a base fluid, achieving tunable spectral absorption by adjusting particle type, concentration, and size. Experimental studies by Liu Dongxue et al. demonstrated that

CPV/T systems using deionized water as the spectral splitter achieved an overall solar energy utilization rate of over 60% [21]. However, absorption-type splitting inevitably converts absorbed photon energy into heat, resulting in unavoidable photon-to-thermal losses for the absorbed bands.

(2) **Reflection-type splitting (dichroic beam splitter):** Utilizes the interference effect of multilayer dielectric films to achieve near-total reflection or transmission for specific wavelength bands. This system adopts this approach—a dichroic beam splitter is placed at the focal point after concentration, with the film system designed to achieve >95% reflectivity for the 400–700 nm band and >90% transmissivity for the 700–2,500 nm band. The key advantage of this scheme is that light energy separation is completed without conversion to other energy forms, allowing both light paths to maintain high optical quality.

(3) **Refraction-type splitting:** Utilizes the dispersion effect of prisms or gratings to spatially separate different wavelengths of light. The linear Fresnel filter-concentrator structure is an innovative recent scheme—the PV-CPT hybrid conversion system based on a linear Fresnel filter-concentrator structure proposed by Wang et al. [22] can simultaneously complete filtering, spectral splitting, and concentration functions within a compact unit, avoiding the additional optical losses and bulk of external beam splitters.

(4) **Holographic-type splitting:** Utilizes the wavelength-selective diffraction characteristics of holographic optical elements (HOEs) to deflect specific wavelength bands to designated directions. This technology is still in the laboratory stage, but theoretically it can achieve more compact and lower-cost spectral splitting.

In the light-guiding transmission stage, fiber optic daylighting technology has received large-scale engineering validation. Yangtze Optical Fiber and Cable's (YOFC) intelligent sunlight guidance system uses rooftop solar trackers to concentrate natural light onto the incident end face of quartz fiber bundles, which is then transmitted through optical fibers indoors and uniformly distributed by diffusers. Eighty-two units of this system have been installed at Xiamen North Railway Station, covering an area of 7,000 m², with an optical efficiency of up to 50%–60% and a transmission distance capability of up to 300 m [23]. This provides a mature engineering reference for the light transmission subsystem of the proposed system.

3.2 Full-Interior-Wall PV Integrating Sphere Power Generation Unit

This is the most innovative core structure of the proposed system. Its design inspiration derives from the integrating sphere (also known as the Ulbricht sphere), widely used in optical measurement.

3.2.1 Physical Model of Integrating Sphere Photon Capture

The integrating sphere is a hollow sphere whose inner wall is coated with a highly diffuse reflective material. According to radiative heat transfer theory, a light beam entering through the entrance aperture strikes a point on the sphere wall, where a portion is absorbed by the wall surface while the remainder undergoes diffuse reflection. The diffusely reflected light follows Lambert's cosine law, uniformly distributing to all other positions on the sphere wall, where it is again partially absorbed and partially reflected. This process repeats cyclically, forming a radiative exchange network between infinitesimal surface elements.

Let the average absorptivity of the sphere wall PV cells for visible light be α (for high-efficiency monocrystalline silicon cells, $\alpha \approx 0.85\text{--}0.90$), the inner surface area of the sphere be S , the entrance aperture area be s , and the aperture ratio be $f = s/S$. In the first incidence, the proportion absorbed by the PV cells is $\alpha(1-f)$ (the first incidence area may also include the aperture edge area not covered by cells). The remaining $1-\alpha(1-f)$ portion undergoes secondary diffuse reflection...

As n approaches infinity, the limit of this geometric series is:

Taking $\alpha = 0.85$ (conservative estimate; PV cell absorptivity for visible light is typically higher) and $f = 0.03$ (aperture ratio $<3\%$), $\eta_{\text{capture}} \approx 0.977$. Taking $\alpha = 0.90$, $\eta_{\text{capture}} \approx 0.987$. That is, theoretically over 97% of incident photons are ultimately absorbed by the PV cells.

More critically, the integrating sphere PV unit is insensitive to incident angle. Due to the highly uniform light field after multiple diffuse reflections inside the sphere, the final photon capture rate is consistent regardless of the angle at which light enters through the aperture. This characteristic significantly reduces the requirement for solar tracking precision—spot shifts caused by tracking errors are rapidly "smoothed out" within the sphere, without forming localized hot spots or efficiency losses.

3.2.2 Sphere Structure and Cell Layout Design

Recommended sphere design parameters: diameter 0.3 m (prototype validation) to 2 m (engineering application), material acrylic or fiberglass-reinforced plastic. The inner wall is lined with fan-shaped cut high-efficiency monocrystalline silicon PV cells (156 mm × 156 mm standard wafer cut into sectors). Cell layout follows a spherical geodesic subdivision pattern to minimize gaps between cells. The cooling air flow channel is embedded behind the cell support structure, with an air intake and exhaust arranged at the bottom and top of the sphere to form a natural or forced convection path.

The sphere also features an optimized inner wall reflective coating design. For areas between cells and the cell surface itself, a broadband high-reflectivity coating (BaSO₄ or PTFE-based diffuse reflective coating, reflectivity >95% in the 400–1,100 nm band) is applied to ensure that unabsorbed photons maintain high reflectivity during subsequent diffuse reflections. The overall structural design needs to achieve an effective aperture ratio of <3% while ensuring sufficient mechanical strength to support the weight of the cells and cooling system.

3.2.3 Integrated Design of Heat Dissipation and Heat Recovery

The integrating sphere PV unit operates indoors, effectively avoiding the dust accumulation problem entirely. Since the PV cells are sealed inside a closed sphere, there is no dust deposition issue at all, eliminating the single largest source of outdoor PV efficiency degradation.

In terms of temperature control, the sphere has a dual advantage. First, since infrared radiation has been spectrally split and directed to the thermal pathway, the PV cells inside the sphere receive only visible light photons, eliminating the thermal load from sub-bandgap photons. Second, the sphere can be placed indoors with active forced-air cooling, maintaining the cell temperature within a controllable range of 35–55°C. The heated exhaust air (70–130°C) is directed to the Kalina cycle evaporator as a medium-temperature heat source, achieving cogeneration of heat and power.

The expected PV performance improvement is summarized as: dust-induced losses 0% (indoor sealed), temperature-induced losses <5% (35–55°C controlled temperature, compared with 14%–28% losses for outdoor PV at 60–80°C), and photon capture advantage +15%–25% (integrating sphere multi-reflection vs. flat single-pass). The combined effect yields an

expected power output per unit cell area 30%–50% higher than conventional ground-mounted outdoor PV.

Economically, although the integrating sphere structure adds cost (sphere shell, reflective coating, air cooling system), this is offset by: (a) the elimination of cleaning costs (the largest component of PV O&M expenditure); (b) reduced land costs (vertical stacking possibility for multi-sphere arrays); and (c) the value of recoverable waste heat. A preliminary cost-benefit estimate is provided in Chapter 7.

3.2.4 Comparison with Existing Concentrated Photovoltaic Receivers

Compared with existing CPV receivers (point-focus triple-junction, linear Fresnel silicon cells, etc.), the integrating sphere PV unit offers three distinctive advantages: (1) Tolerance for tracking errors—the integrating sphere naturally homogenizes the light field, requiring only coarse tracking; (2) No localized hot spots—multi-reflection uniformly distributes concentrated light energy, avoiding the cooling challenges of conventional CPV cells under hundreds of suns; (3) Cogeneration capability—the cooling airflow carries recoverable heat at a useful temperature level. Its disadvantage is the relatively complex sphere structure and the additional cost of multi-junction cells if ultra-high efficiency is pursued.

The key comparison data are summarized in Table 3 (inserted after the sensitivity analysis section). Currently, no commercial or laboratory-scale integrating sphere PV system has been reported, making this a truly original structural concept.

3.3 Supercritical CO₂ Brayton Cycle

3.3.1 Technical Principle and Configuration Selection

The supercritical CO₂ (sCO₂) Brayton cycle is a closed thermodynamic cycle using CO₂ above its critical point (31°C, 7.38 MPa) as the working fluid. Near the critical point, CO₂ exhibits liquid-like density (approximately 468 kg/m³) and gas-like low viscosity, significantly reducing compression work, enabling higher cycle efficiencies than conventional steam Rankine cycles at equivalent temperature levels, while the turbomachinery is more compact.

Among various sCO₂ cycle configurations, the recompression cycle architecture is considered the most efficient and is currently the mainstream design choice for commercial

demonstration projects. Its core feature is the use of two compressors and two recuperators to reduce external heat rejection, achieving a thermal efficiency of approximately 44%–49% at 700°C turbine inlet temperature.

In this system, the sCO₂ cycle serves as the high-temperature power generation unit, receiving infrared heat from the spectral splitting stage. Key design parameters: turbine inlet temperature 550–700°C, turbine inlet pressure 20–25 MPa, exhaust temperature 250–350°C. The sCO₂ exhaust heat is not discharged but serves as the primary heat source for the subsequent Kalina cycle, embodying the cascade utilization principle.

3.3.2 Global Demonstration Projects and Technology Progress

sCO₂ power cycle technology has advanced rapidly in recent years. Key global demonstration milestones include: the U.S. Department of Energy's STEP (Supercritical Transformational Electric Power) 10 MWe pilot facility achieved 10 MW full-load operation [25]; China's first commercial sCO₂ power generation unit, "SuperCarbon-1" (Chaotan-1), was commissioned in 2025, with an initial 1 MW scale [26]; Beijing Frontier Power won the bid for Xi'an Jiaotong University's 100 kW-class sCO₂ integrated unit project in 2026 [27]; and the Institute of Engineering Thermophysics, Chinese Academy of Sciences, reported progress on a 2.1 MW radial inflow sCO₂ turbine [28]. These projects indicate that sCO₂ technology is entering pre-commercial maturity, with current costs of approximately 5 million CNY/MW.

For the proposed system, the sCO₂ cycle is assessed at Technology Readiness Level (TRL) 6–7 (system/subsystem prototype demonstrated in an operational environment), with an estimated 2–3 years to commercial maturity.

3.4 Kalina Cycle

3.4.1 Operating Principle and Thermodynamic Advantages

The Kalina cycle, proposed by Alexander Kalina in 1984 [16], uses an ammonia-water mixture as the working fluid to replace the pure water/steam working fluid of the conventional Rankine cycle. The key thermodynamic advantage of the ammonia-water mixture lies in its non-isothermal boiling and condensation characteristics.

In this system, the Kalina cycle serves as the medium-temperature power generation stage, receiving heat from three sources: (1) sCO₂ exhaust afterheat (250°C→150°C), (2)

integrating sphere PV cooling air (70–130°C), and (3) AI data center liquid cooling return water (35–50°C, preheated to 80–100°C). The multi-source confluence characteristic is ideally suited to the Kalina cycle's non-isothermal evaporation, as the variable-temperature heat source cooling curve from the three streams closely matches the working fluid's temperature-rise evaporation curve, minimizing the pinch point exergy loss inherent in conventional heat exchange.

3.4.2 Configuration Evolution and Applicable Temperature Range

Various Kalina cycle configurations have been developed for different application scenarios [29]. The most relevant for this system are: the KCS11 configuration (suitable for low-temperature heat sources in the 80–150°C range) and the KCS34 configuration (suitable for medium-temperature heat sources of 150–350°C). Given the multi-source confluence characteristics of this system (heat source temperature range 35–350°C), the KCS34 configuration with a distillation-condensation sub-cycle (DCSC) is recommended, as it can more flexibly accommodate the multi-point heat input requirements.

3.4.3 Global Commercial Operation Validation

The Kalina cycle has over 27 years of commercial operating history. The first commercial Kalina cycle power plant (Husavik, Iceland, 2 MW, geothermal heat source) entered service in 2000 and continues to operate reliably. As of 2025, over 20 Kalina cycle power plants have been deployed globally, including Japan's largest geothermal Kalina plant at the Onikobe geothermal field (5 MW commissioned 2023) and several biomass and industrial waste heat recovery applications across Germany, the United States, and South Korea. This proven track record validates the Kalina cycle's reliability and durability for this proposed system.

3.5 AI Data Center Liquid Cooling Waste Heat Recovery

3.5.1 Liquid Cooling Technology Routes and Return Water Characteristics

3.5.2 Economic Analysis of Solar Preheating

3.5.3 Phase-Change Emulsion and High-Temperature Thermal Oil Alternatives

3.6 Vortex Tube Grade Restructuring and Radiative Cooling

3.6.1 Physical Model of the Vortex Tube

3.6.2 Physical Mechanism of Radiative Cooling and Material Advances

4. Full-Chain Efficiency Model and Exergy Analysis

4.1 Model Assumptions and Boundary Conditions

This efficiency model is based on the following assumptions (numbered individually for reviewer reference):

A1: Solar radiation adopts the AM1.5G standard spectrum, total irradiance 1,000 W/m², DNI $\geq 1,800$ kWh/m²·yr, corresponding to Class I solar energy resource regions.

A2: Spectral splitting uses solid-state dichroic beam splitters, visible light reflection efficiency 90%, infrared transmission efficiency 90%.

A3: Optical fiber light-guiding system transmission efficiency 85% (including coupling losses, fiber attenuation, and terminal scattering), transmission distance ≤ 300 m.

A4: Integrating sphere PV unit aperture ratio $< 3\%$, PV cells are high-efficiency monocrystalline silicon (efficiency 22.5%), cell absorptivity 85%, photon capture rate $> 95\%$.

A5: sCO₂ Brayton cycle adopts recompression configuration, turbine inlet temperature 550–700°C, cycle efficiency taken as 44% (conservative; up to 49% at 700°C), turbine outlet temperature 250–350°C.

A6: Kalina cycle heat source temperature 80–350°C, KCS11/KCS34 configuration, combined thermal-to-electric efficiency 12%.

A7: AI data center installed capacity 22.5 MW, liquid cooling return water temperature 35–50°C, waste heat ratio 40% (i.e., waste heat power 9 MW).

A8: Vortex tube uses system residual pressure airflow as gas source, cold end temperature drop 30°C, hot end temperature rise 20°C, equivalent recovered electricity 0.8 units.

A9: Radiative cooling film temperature drop 18°C, PV efficiency relative improvement 8%, net gain approximately 1 unit (baseline normalized).

A10: System parasitic loads (pump work, compression work, trackers, etc.) account for 15% of total power generation output.

4.2 Independent Efficiency Equations for Each Subsystem

Spectral splitting and light-guiding subsystem:

sCO₂ Brayton cycle subsystem:

Kalina cycle subsystem (three-source confluence):

Total system power generation:

Net output:

Full-chain net efficiency:

In the above calculations, PV cell efficiency and sCO₂ cycle efficiency were both taken as conservative values. Using more optimistic values (PV cell efficiency 25%, sCO₂ cycle efficiency 49%), the system net efficiency can reach approximately 37.1% (as discussed in the previous chapter). Based on this, three estimation scenarios are constructed:

4.3 Sensitivity Analysis of Key Parameters

Table 2 Sensitivity Analysis of Key Parameters (Effect of ±20% Variation on Net Efficiency)

Scenario	PV Efficiency	sCO ₂ Efficiency	Kalina Efficiency	System Net Efficiency
Conservative	22.5%	44%	10%	31.5%
Baseline	25%	44%	12%	32.4%
Optimistic	25%	49%	14%	37.1%

The sensitivity analysis indicates that sCO₂ cycle efficiency and PV cell efficiency are the most critical parameters for system performance—these two subsystems are also the most rapidly developing areas of technology. With the commercialization of sCO₂ technology and the advancement of perovskite/crystalline silicon tandem cell efficiency toward 30%, the system net efficiency is expected to exceed 40% within 10 years.

Table 3 Detailed Sensitivity Analysis (±20% Variation on System Net Efficiency)

Parameter	−20% Impact	+20% Impact	Sensitivity Rating
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Splitting Efficiency (Baseline 0.9)	−3.2%	+2.1%	High
Fiber Transmission Efficiency (Baseline 0.85)	−3.8%	+3.8%	High
Integrating Sphere Capture Rate (Baseline 0.95)	−2.5%	+1.6%	Medium-High
PV Cell Efficiency (Baseline 0.25)	−6.5%	+6.5%	Very High
sCO ₂ Cycle Efficiency (Baseline 0.44)	−6.8%	+6.8%	Very High
Kalina Cycle Efficiency (Baseline 0.12)	−1.9%	+1.9%	Medium

4.4 Exergy Analysis Framework

5. AI Data Center Synergy Mechanism

5.1 Mathematical Description of Thermodynamic Coupling

5.2 Optimization Model for "Compute-Heat-Electricity" Dynamic Dispatch

Optimization objective function:

Constraints:

5.3 Digital Twin Architecture Design

Physical layer: The actual operating hardware—light collection trackers, beam splitters, integrating sphere PV units, sCO₂ units, Kalina units, data center liquid cooling system, OTEC system, etc.

Perception layer: A sensor network deployed at key nodes, collecting parameters such as temperature, pressure, flow rate, irradiance, voltage, and current in real time, with a sampling frequency of 1–10 Hz.

Model layer: Multi-physics coupled models established using tools such as COMSOL Multiphysics, including an optical ray-tracing model, thermodynamic cycle model, heat exchanger distributed parameter model, and data center thermal distribution model. Model parameters are calibrated using pilot experiment data.

Decision layer: An optimal scheduling strategy engine trained by reinforcement learning, which outputs control commands for each subsystem based on real-time perception data and electricity price/weather predictions.

Application layer: Human-machine interface, alarm management, performance reports, carbon asset management, etc.

6. Multi-Scenario Deployment Strategy

6.1 Four-Dimensional Clustering-Based Scenario Classification

Scenario 1: Tropical remote islands (high temperature and humidity, good solar resources, large seawater temperature differences, weak grid infrastructure)

6.2 Tropical Remote Island Solution

6.3 Coastal AI Intelligent Computing Center Cluster Solution

6.4 Large Industrial Complex Solution

7. Economic Analysis

7.1 Full Life-Cycle Cost Model

Initial investment (CAPEX):

Annual operating cost (OPEX): labor, maintenance, insurance, consumables, etc., estimated at 2%–2.5% of CAPEX, approximately 10–12.5 million CNY/year. In addition, grid electricity purchases (nighttime or cloudy day makeup) are temporarily estimated at 1 million CNY/year. Total OPEX approximately 12 million CNY/year.

Table 4 System CAPEX Breakdown

Subsystem	Investment (10,000 CNY)	Share	Unit Price Reference
Light Collection & Splitting	12,000	24%	Fresnel lens field ~800 CNY/m², custom dichroic splitter
Integrating Sphere PV System	15,000	30%	Monocrystalline Si cells + sphere structure + cooling
sCO ₂ + Kalina Thermal System	10,000	20%	100 kW-class sCO ₂ integrated unit ~5M CNY/MW
AI Data Center Module	8,000	16%	Liquid-cooled servers incl.

			cooling ~4M CNY/MW
OTEC + Desalination Module	3,000	6%	OTEC ~3M CNY/MW, LT-MED ~0.8M CNY/(t·h)
Balance of System & Installation	2,000	4%	Pipelines, electrical, civil works
Total CAPEX	50,000	100%	—

7.2 Full-Spectrum Revenue Estimation

Revenue item 1—Computing services: $20 \text{ MW} \times 8,760 \text{ h} \times 80\% \text{ utilization} \times 1.5 \text{ CNY/TOPS} \cdot \text{h}$ (based on current AI inference service pricing), the annual revenue is approximately 420 million CNY. This estimate does not account for the risk of computing price decline with technological progress; a $\pm 30\%$ fluctuation range is recommended for sensitivity analysis.

Revenue item 2—Self-supplied electricity savings: $10 \text{ MW} \times 1,800 \text{ equivalent full-load hours} \times 0.8 \text{ CNY/kWh}$ (commercial/industrial electricity price), annual electricity cost savings of approximately 14.4 million CNY.

Revenue item 3—Freshwater sales: $311 \text{ t/h} \times 1,800 \text{ h} \times 10 \text{ CNY/t}$, annual revenue approximately 5.6 million CNY.

Revenue item 4—Waste heat heating/cooling: annual revenue approximately 2 million CNY (replacing electricity costs for northern heating or southern cooling).

Revenue item 5—Carbon reduction benefits: assuming the system reduces CO₂ emissions by approximately 50,000 t/year, at a carbon price of 100 CNY/tCO₂, carbon revenue is approximately 5 million CNY/year (this benefit is partially reflected in the self-supplied electricity savings, posing a double-counting risk; a 50% discount is recommended, i.e., 2.5 million CNY/year).

Total annual revenue: approximately 442 million CNY (approximately 440 million CNY after deducting double-counted carbon benefits).

7.3 Sensitivity Analysis and Scenario Analysis

Optimistic scenario (computing price stable or rising, equipment cost down 20%): NPV >1 billion CNY, IRR >35%

Baseline scenario (computing price moderately declining, equipment cost down 10%): NPV approximately 500–700 million CNY, IRR approximately 20%–25%

Pessimistic scenario (computing price halved, equipment cost unchanged): NPV approximately 100–200 million CNY, IRR approximately 8%–12%

7.4 Comprehensive Comparison with Conventional Solutions

Table 5 Comprehensive Comparison with Conventional Solutions

Indicator	This System	Conventional PV + Standalone Data Center
PV Annual Effective Generation Hours	1,800 h (Indoor, constant temp, no degradation)	1,200–1,500 h (Outdoor, dust + high-temp degradation)
PV Full Life-Cycle Degradation Rate	<0.1%/year (Indoor sealed)	0.5%–0.8%/year (Outdoor aging)
Data Center PUE	<1.10	1.25–1.40
Land Use Efficiency (kW/m ²)	0.3–0.5 (Multi-sphere stacking)	0.15–0.20 (Flat layout)
Water Consumption (L/kWh)	0–0.02 (Seawater cooling, no freshwater)	1.5–2.5 (Evaporative cooling tower)
Waste Heat Utilization Rate	5%–12% of total power	0%
Carbon Footprint (g CO ₂ /kWh)	<50 (Full life cycle)	300–500 (Grid-dependent)
Energy Self-Sufficiency Rate	>80%	<30% (Data center relies on grid)
Multi-Product Output	Electricity, freshwater, cooling, minerals	Electricity only

8. Challenges, Limitations, and Future Directions

8.1 Technical Limitations

Scaling up the integrating sphere PV unit. The photon capture rate of the integrating sphere remains unchanged as the sphere diameter increases (depending only on aperture ratio and cell absorptivity), but heat dissipation difficulty and manufacturing complexity increase significantly. For large-diameter spheres (>2 m), issues such as inner-wall cell splicing, lead-through sealing, and cooling uniformity require dedicated engineering development. Preliminary analysis indicates that the preferred approach for a single integrating sphere unit is a 0.5–1 m diameter module, with multi-unit arrays achieving megawatt-scale capacity through parallel stacking.

The multi-layer vertical stacking design (multi-sphere, multi-layer integrated framework) can further increase land-use efficiency.

Frequent start-stop fatigue of sCO₂ cycle. Solar energy has diurnal and weather fluctuation characteristics. If the sCO₂ unit is designed for daytime operation and nighttime shutdown, it will face thermal fatigue issues from frequent start-stop cycles. Turbine blades and recuperators may accumulate low-cycle fatigue under repeated high-temperature to cooling cycles. Solutions include: (a) configuring molten salt thermal storage to smooth daytime heat input and maintain nighttime operation at reduced load; (b) using a bypass system to keep the sCO₂ unit in a hot standby state during short cloud-cover intervals, reducing thermal cycling.

Biofouling and corrosion of OTEC deep seawater pipes. Deep seawater pipes (typically >1 m diameter, >1,000 m length) are immersed in seawater for extended periods, facing dual threats from biofouling (barnacles, shellfish, etc.) and electrochemical corrosion. Existing solutions include copper alloy coatings (anti-fouling), cathodic protection (anti-corrosion), and periodic ROV cleaning, but all increase operating costs.

Efficiency degradation of vortex tube under variable operating conditions. Vortex tube performance is sensitive to inlet pressure, temperature, and cold flow fraction. Under variable operating conditions (where solar irradiance fluctuations cause changes in compressed gas source parameters), the hot-cold end temperature difference may fluctuate significantly. It is recommended to install a pressure-stabilizing tank and pressure-regulating valve upstream of the vortex tube to stabilize its operating point.

8.2 Competitive Analysis of Alternative Technology Pathways

Perovskite/crystalline silicon tandem PV cells. The current laboratory efficiency of perovskite/c-Si tandem cells has exceeded 33%. If mass production is achieved within the next 10 years with controllable costs, PV efficiency could see a substantial leap. However, tandem cells do not solve the dust and high-temperature problems—if still placed outdoors, the efficiency advantage will be partially offset by degradation. The indoor integrating sphere solution of this system is compatible with any PV technology, including future ultra-high-efficiency tandem cells. Once installed inside the sphere, tandem cells would benefit from both dust-free and temperature-controlled conditions.

Direct sCO₂ cooling of data centers. Some researchers have proposed using sCO₂ directly as the data center cooling working fluid, allowing it to evaporate from the supercritical state at the chip heat source to drive a turbine for power generation. This scheme eliminates the intermediate liquid cooling stage, theoretically achieving higher heat recovery efficiency, but faces significant engineering challenges including sCO₂ compatibility with electronic components (high pressure, corrosion) and leakage risk.

Small Modular Nuclear Reactors (SMRs) powering data centers. The concept of SMRs as base-load power sources for data centers is gaining attention (e.g., Microsoft's partnership with Helion Energy). The advantage of SMRs is their complete independence from weather conditions; disadvantages include site selection constraints, low public acceptance, and uncertain nuclear waste disposal and decommissioning costs. This system and SMRs are not mutually exclusive—in high-latitude regions or locations with persistently poor solar resources, an SMR could replace or supplement the solar thermal component while retaining the cascade architecture.

8.3 Policy and Market Risks

Carbon price volatility risk. Carbon reduction benefits are one of the system's ancillary revenue sources. The national carbon market price is currently approximately 60–80 CNY/tCO₂. If carbon prices decline or carbon quota policies are adjusted in the future, the system's carbon revenue expectations will be affected. However, low-carbon development is a long-term structural trend, making a significant decline in carbon prices unlikely.

Computing service price decline risk. AI computing service prices are influenced by three factors: technological progress, competitive landscape, and market demand. Large-model inference costs are declining at a rate exceeding 50% per year (e.g., OpenAI's token prices have been reduced year after year). If computing becomes a "commodity" in the future with significantly compressed profit margins, the system's economic returns will be severely impacted. As a countermeasure, the system's multi-product structure (electricity, freshwater, cooling, carbon benefits) provides diversified revenue streams, reducing dependence on a single computing service income source.

Technology lock-in risk. This system is highly integrated, involving multiple frontier technologies. If any one technology (such as sCO₂ or Kalina) fails to achieve the

expected commercial maturity, the overall techno-economics of the system will be affected. It is recommended to maintain flexibility in the technology pathway during the early stage—for example, the Kalina cycle can be downgraded to ORC (with reduced efficiency but still operable), and the sCO₂ cycle can be replaced by a conventional steam Rankine cycle as a fallback.

8.4 Future Research Directions

10-times scale-up experiment of integrating sphere PV: scaling up from $\phi 0.3$ m prototype to $\phi 1\text{--}2$ m engineering prototype, verifying the scale invariance of photon capture efficiency and the effectiveness of the scaled-up cooling solution.

Long-term compatibility testing of high-temperature liquid cooling media: conducting ≥ 1 year phase-change emulsion/thermal oil circulation tests in an actual data center environment, evaluating corrosion and aging effects on server components (pipes, connectors, seals).

Digital twin model calibration based on measured data: using pilot plant operational data to calibrate the multi-physics coupled model parameters, improving the prediction accuracy of the digital twin.

Economic modeling of computing-energy market coupling mechanisms: establishing a game-theoretic model of the "computing market–electricity market–carbon market" to analyze the price linkage relationships among the three and the system's optimal bidding strategy.

Economic feasibility verification of valuable metal extraction from concentrated brine: operating lithium and uranium adsorption-separation devices at pilot scale to obtain actual yield and cost data, verifying the commercial feasibility of mineral extraction as an auxiliary revenue source for the system.

9. Conclusion

First, the spectral splitting + full-interior-wall integrating sphere PV architecture solves three major pain points of conventional PV from physical principles: dust accumulation, high-temperature degradation, and insufficient single-pass photon absorption. Theoretical analysis shows that the photon capture rate of the integrating sphere PV unit can exceed 97%, with an expected power output 15%–25% higher than

an equal-area flat-mounted PV array. This is a structural innovation based on mature optical principles.

Second, the series structure of the sCO₂ Brayton cycle + Kalina cycle achieves full-temperature-range cascade utilization from 700°C high temperature down to 80°C low temperature, with sound thermodynamic matching. The sCO₂ cycle is on the eve of commercialization (the world's first commercial unit has been commissioned), and the Kalina cycle has over 27 years of commercial operating validation. The combination of the two achieves a cascade effect that neither can achieve alone.

Third, the injection of AI data center liquid cooling return water realizes the paradigm shift of "computing power equals thermal power," transforming the data center from a net electricity consumer into a thermal contribution unit. This coupling has synergistic effects in four dimensions—temperature, time, space, and economics—and is expected to increase data center waste heat recovery to 5%–12% of total power consumption.

Fourth, the vortex tube grade restructuring and radiative cooling films form a "zero-energy-consumption cold skeleton," achieving end-stage cooling and grade upgrading without additional power consumption. Both technologies are supported by existing commercial products.

Fifth, the all-factor solution for coastal scenarios (OTEC base load + seawater desalination + mineral extraction) provides an energy self-sufficiency solution "from sunlight and seawater to everything" for more than 100,000 islands and coastal AI clusters globally.

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Appendix: Glossary

中文术语	英文术语	缩写
光谱分频	Spectral Splitting	—
积分球	Integrating Sphere	—
超临界二氧化碳	Supercritical Carbon Dioxide	sCO ₂
布雷顿循环	Brayton Cycle	—
卡琳娜循环	Kalina Cycle	—
有机朗肯循环	Organic Rankine Cycle	ORC
海洋温差能	Ocean Thermal Energy Conversion	OTEC
低温多效蒸馏	Low-Temperature Multi-Effect Distillation	LT-MED
涡流管	Vortex Tube	—
辐射制冷	Radiative Cooling	—
温差发电	Thermoelectric Generator	TEG
电能利用效率	Power Usage Effectiveness	PUE
直接法向辐照	Direct Normal Irradiance	DNI
平准化能源成本	Levelized Cost of Energy	LCOE
平准化淡水成本	Levelized Cost of Water	LCOW
碳边境调节机制	Carbon Border Adjustment Mechanism	CBAM
服务等级协议	Service Level Agreement	SLA
净现值	Net Present Value	NPV
内部收益率	Internal Rate of Return	IRR
焓	Exergy	—